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Lab section B01

ENCM 335 - LAB 1

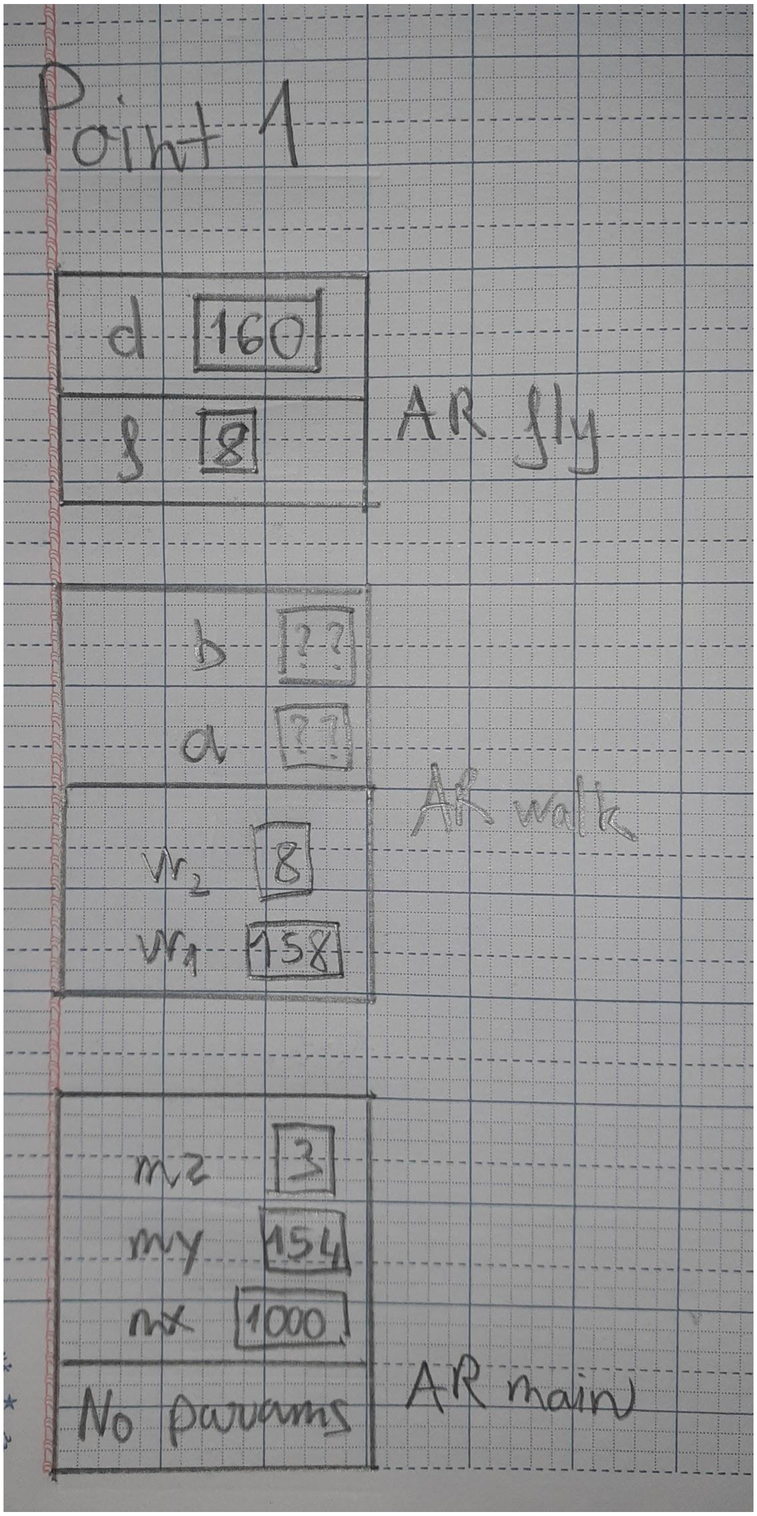
Exercise A:

lab2exA.c

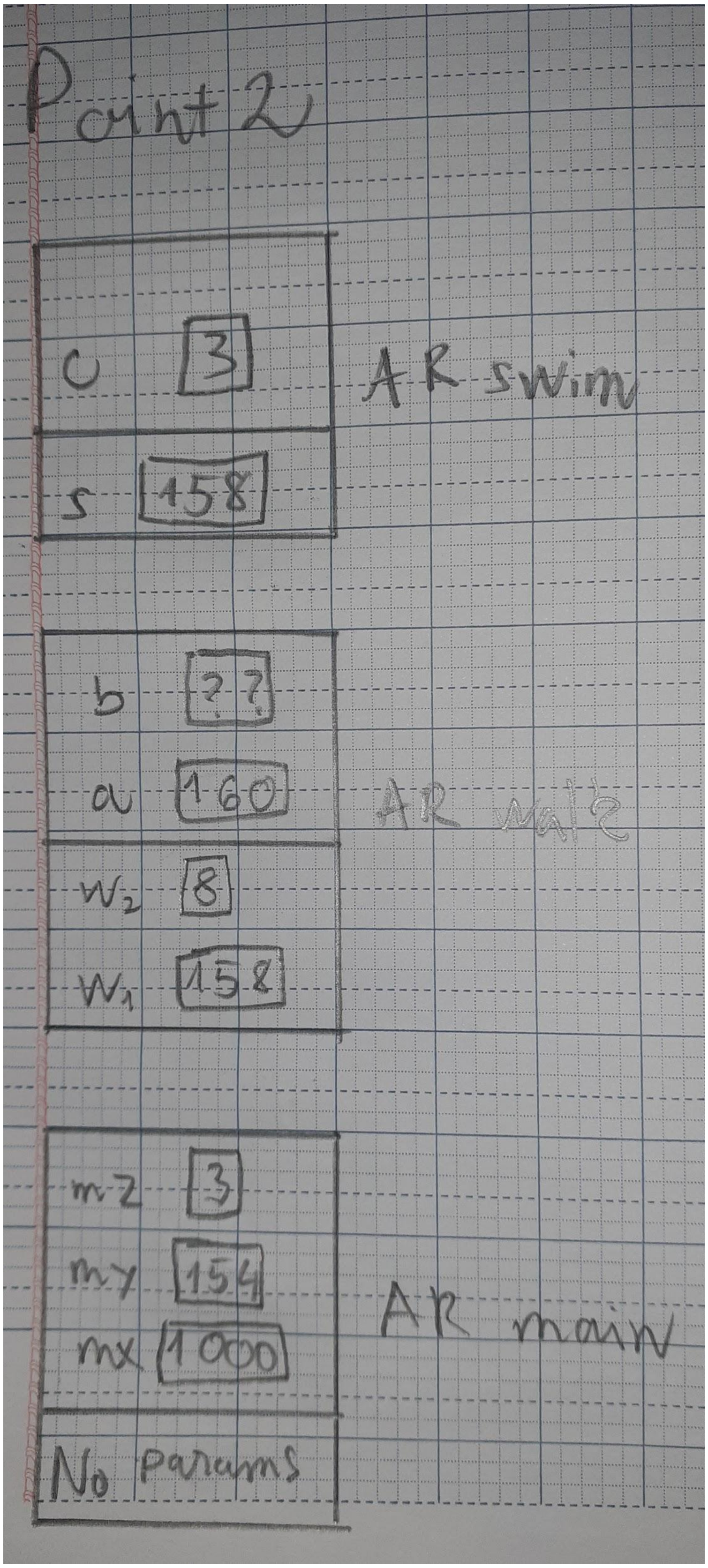
```c
// ENCM 335 Fall 2021 Lab 2 Exercise A

#include <stdio.h>
#include <stdlib.h>

int main(void)
{
  int j, k, input_count;
  printf("Please enter two integers, separated by a space:\n");
  input_count = scanf("%d %d", &j, &k);

  // If scanf didn't get two values, quit the program.
  if (input_count != 2) {
    printf("Sorry, that input was not valid.  I'm quitting!\n");
    exit(1);
  }

  // Echo the input.
  printf("The values read were %d for j and %d for k.\n", j, k);

  // Add code below this comment to complete the exercise ...
  if ((j<0) && (k<0)){
      printf("Both j and k are negative");
  }
  else if ((j<0) && (k>0)){
      printf("j is negative but k is not");
  }
  else if ((j>0) && (k<0)){
      printf("k is negative but j is not");
  }
  else if ((j>=0) && (k>=0)){
      printf("Neither j nor k is negative");
  }

  return 0;
}
```

Exercise B:

Point 1:



Point 2:


Point 2
c 3
s 158
AR swim
b ??
a 160
AR walk
w_2 8
w_1 158
m-z 3
my 154
mx 1000
AR main
No params

Exercise D:

Code:

lab2exD.c

```c
// ENCM 335 Fall 2021 Lab 2 Exercise D

#include <stdio.h>


int is_valid_time(int hour, int min, int sec)
{
// Returns 1 if the parameters describe a valid time of day
// between 00:00:00 and 23:59:59.
// Returns 0 otherwise.
  if ((hour>=0)&&(hour<24)){
      if ((min>=0)&&(min<60)){
          if ((sec>=0)&&(sec<60)){
              return 1;
          }
      }
  }

  return 0;


}

```

lab2exD.c

```c

void call_is_valid_time(int hour, int min, int sec)
// Prints parameters within in a message stating whether
// the parameters make sense together as a time of day.

{
  // Note %02d says "print an int with at least 2 digits,
  // and fill with leading zeros if necessary.
  printf("%02d:%02d:%02d ", hour, min, sec);
  if (is_valid_time(hour, min, sec))
    printf("makes");
  else
    printf("does not make");
    printf(" sense as a time of day.\n");

}

void describe(int h, int min, int s)
{
// Assumes that the parameters describe a valid time of day.
// Prints the time and a description of the time using terms
// such as "wee hours", "morning", etc., as describe in the Lab 1
// instructions.

printf("%02d:%02d:%02d is in the ", h, min, s);

```

```c

if (h<5){
     printf("wee hours.\n");
}
else if (h==5){
     if (min<=29){
         printf("wee hours.\n");
     }
     else {
         printf("morning.\n");
     }
}
else if (h<=11){
     printf("morning.\n");
}
else if (h==12){
     printf("afternoon.\n");
}
else if (h<17){
     printf("afternoon.\n");
}
else if (h==17){
```

lab2exD.c

```c
else if (h==17){
    if (min<=30){
        if (s<=19){
            printf("afternoon.\n");
        }
        else {
            printf("evening.\n");
        }
    }
    else {
        printf("evening.\n");
    }
}

else if (h<23){
    printf("evening.\n");
}
else if (h==23){
    if (min==0){
        if(s==0){
            printf("evening.\n");
        }
        else {
            printf("late night.\n");
        }
    }
    else {
        printf("late night.\n");
    }

```

```c

}
else
    printf("late night.\n");

}



int main(void)
{
  call_is_valid_time(-1, 0, 0);
  call_is_valid_time(0, -1, 0);
  call_is_valid_time(0, 0, -1);

  call_is_valid_time(24, 0, 0);
  call_is_valid_time(0, 60, 0);
  call_is_valid_time(0, 0, 60);

  call_is_valid_time(0, 0, 0);
  call_is_valid_time(23, 59, 59);

  printf("\n");

  describe(0, 0, 0);
  describe(3, 12, 27);
  describe(5, 29, 59);
  describe(5, 30, 0);
```

```c
  describe(5, 30, 0);
  describe(10, 3, 58);
  describe(11, 59, 59);
  describe(12, 0, 0);
  describe(3, 0, 0);
  describe(17, 30, 19);
  describe(17, 30, 20);
  describe(17, 30, 21);
  describe(22, 59, 59);
  describe(23, 0, 0);
  describe(23, 0, 1);
  describe(23, 59, 59);

  return 0;
}
```

Output:

~/ENCM_335/lab2

```
monsi@LAPTOP-M9ONH6KV ~
$ cd ENCM_335/lab2

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab2
$ gcc -Wall lab2exD.c

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab2
$ ./a.exe
-1:00:00 does not make sense as a time of day.
00:-1:00 does not make sense as a time of day.
00:00:-1 does not make sense as a time of day.
24:00:00 does not make sense as a time of day.
00:60:00 does not make sense as a time of day.
00:00:60 does not make sense as a time of day.
00:00:00 makes sense as a time of day.
23:59:59 makes sense as a time of day.

00:00:00 is in the wee hours.
03:12:27 is in the wee hours.
05:29:59 is in the wee hours.
05:30:00 is in the morning.
10:03:58 is in the morning.
11:59:59 is in the morning.
12:00:00 is in the afternoon.
03:00:00 is in the wee hours.
17:30:19 is in the afternoon.
17:30:20 is in the evening.
17:30:21 is in the evening.
22:59:59 is in the evening.
23:00:00 is in the evening.
23:00:01 is in the late night.
23:59:59 is in the late night.

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab2
$ |
```

Exercise E:

lab2exD.c ✖ lab2exE.c ✖

```c
// ENCM 335 Fall 2021 Lab 2 Exercise E

#include <math.h>
// The above directive will read many function prototypes, including ...
//
//     double sin(double x);
//
// for the sine function.  It's assumed that the units for the argument
// value are radians, not degrees.

// Note to Linux users: You might need to build the executable like this ...
//   gcc -Wall lab1exF.c -lm
// ... in order to link in the sin function.  Mac and Cywgin users won't
// have to put -lm at the end of the command.

#include <stdio.h>

// For now, just trust that the following #define directives properly
// sets up a useful constant.  How it works will be explained
// later in the course.
#define PI 3.14159265358979323846


double deg2rad(double degrees);

int main(void)
{
    int row,i,j;

    printf("degrees");
```

lab2exD.c ✖ lab2exE.c ✖

```c


double deg2rad(double degrees);

int main(void)
{
    int row,i,j;

    printf("degrees");
    for (i=0;i<10;i++){
        printf("      +%d",i);
    }
    printf("\n");
    for (row = 0; row < 90; row += 10)  {
        printf("%7d ", row);
        for (j=0;j<10;j++){
            printf("%.4f ",sin(deg2rad(row+j)));

        }
        printf("\n");
    }
    return 0;
}

double deg2rad(double degrees)
{
    return (PI / 180.0) * degrees;
}

```

Output:

~/ENCM_335/lab2

```
monsi@LAPTOP-M9ONH6KV ~
$ cd ENCM_335/lab2

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab2
$ gcc -Wall lab2exE.c

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab2
$ ./a.exe
degrees      +0     +1     +2     +3     +4     +5     +6     +7     +8     +9
      0 0.0000 0.0175 0.0349 0.0523 0.0698 0.0872 0.1045 0.1219 0.1392 0.1564
     10 0.1736 0.1908 0.2079 0.2250 0.2419 0.2588 0.2756 0.2924 0.3090 0.3256
     20 0.3420 0.3584 0.3746 0.3907 0.4067 0.4226 0.4384 0.4540 0.4695 0.4848
     30 0.5000 0.5150 0.5299 0.5446 0.5592 0.5736 0.5878 0.6018 0.6157 0.6293
     40 0.6428 0.6561 0.6691 0.6820 0.6947 0.7071 0.7193 0.7314 0.7431 0.7547
     50 0.7660 0.7771 0.7880 0.7986 0.8090 0.8192 0.8290 0.8387 0.8480 0.8572
     60 0.8660 0.8746 0.8829 0.8910 0.8988 0.9063 0.9135 0.9205 0.9272 0.9336
     70 0.9397 0.9455 0.9511 0.9563 0.9613 0.9659 0.9703 0.9744 0.9781 0.9816
     80 0.9848 0.9877 0.9903 0.9925 0.9945 0.9962 0.9976 0.9986 0.9994 0.9998

monsi@LAPTOP-M9ONH6KV ~/ENCM_335/lab2
$
```